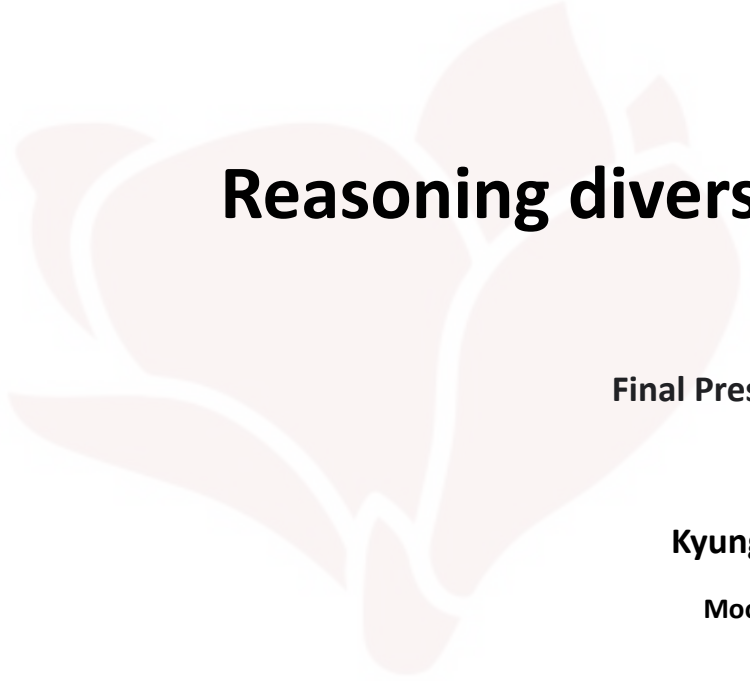




Reasoning diversity for Self-distillation

Final Presentation, 2026. 6. 4

Kyung Hee University

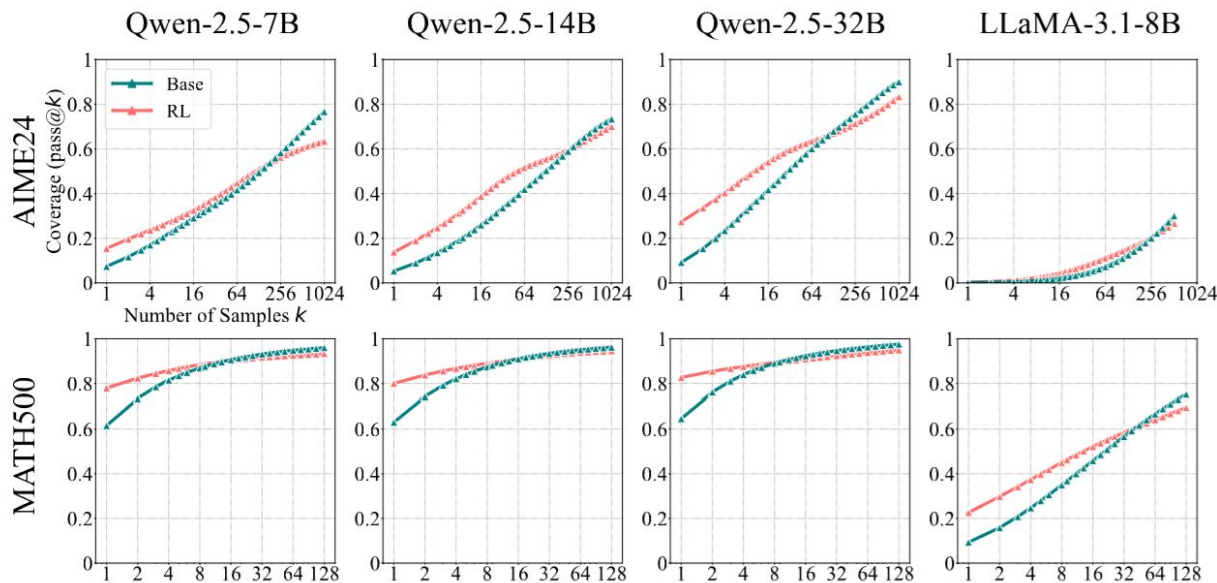
Moon, Ahn and Jeong

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3. Discussions
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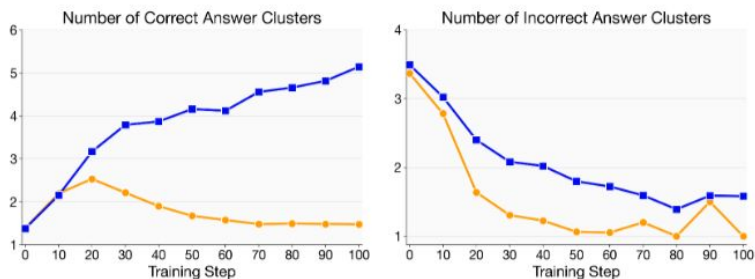
Introduction

- **Problem coverage decreases after RL post-training**
 - The hope of self-improvement has cracked
 - RLVR increases sampling efficiency, not its reasoning capability

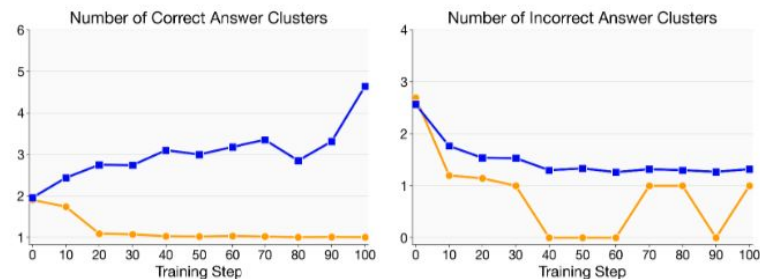


Introduction

- **Strategy Collapse** during standard RLVR
 - RL trained model sticks with the single strategy
 - However, we may want our models to provide diverse solutions (e.g., DP)



Multi-digit Multiplication



Polynomial Solving

GRPO Poly-EPO

Introduction

- Why does the RLVR narrow the coverage?
 - During training, the parameters of the model changes
 - For unsolved problems, **we might lose the capability** of sampling the correct solution forever

$$\theta_{t+1} = \theta_t + \alpha \hat{g}$$

Introduction

- Why does the RLVR narrow the coverage?
 - During training, the parameters of the model changes
 - For unsolved problems, **we might lose the capability** of sampling the correct solution forever
 - This is unavoidable, then **how can we re-discover** the solution that could have originally been sampled?

$$\theta_{t+1} = \theta_t + \alpha \hat{g}$$

Introduction

- Why does the RLVR narrow the coverage?
 - During training, the parameters of the model changes
- We need a well-grounded exploration method
 - The agent needs interact with the environment **with plausible strategies**, even though the current trial may be incorrect
 - However, **naive exploration is impossible** due to its large search space

Method

- Reinforcement Learning via Self-Distillation
 - LLM agents can receive text feedback even though they fail

SDPO: Self-Distillation Policy Optimization

1. Question x

Write a python function that returns all numbers from 1 to n. Answer briefly.

3. Feedback f

Don't include n.

2. Answer $y \sim \pi_\theta(\cdot | x)$

```
```python
def numbers_up_to_n(n):
 return list(range(1, n + 1))
```
```

4. Credit assignment by **self-teacher** $\pi_\theta(y | x, f)$

Method

- Reinforcement Learning via Self-Distillation
 - LLM agents can receive text feedback even though they fail
 - They may rectify and **improve their failed trajectory** via feedback
 - Sets the stage for **exploration within a reasonable** search space

SDPO: Self-Distillation Policy Optimization

1. Question x

Write a python function that returns all numbers from 1 to n. Answer briefly.

3. Feedback f

Don't include n.

2. Answer $y \sim \pi_\theta(\cdot | x)$

```
```python
def numbers_up_to_n(n):
 return list(range(1, n + 1))
```
```

4. Credit assignment by **self-teacher** $\pi_\theta(y | x, f)$

Method

- SDPO Loss: Reverse KL Divergence
 - Maximize negative reverse KL (=minimize regular reverse KL)

$$\max_{\text{vocabs}} - \sum_{t=1}^{|y|} \mathbb{E}_{\hat{y}_t \sim \pi_{\theta}(\cdot | x, y_{<t})} \left[\text{KL}(\pi_{\theta}(\cdot | x, y_{<t}) || \text{stopgrad}(\pi_{\theta}(\cdot | x, f, y_{<t}))) \right]$$

student self-teacher

Method

- SDPO Loss: Reverse KL Divergence
 - Increase the **probability of the vocab(=action)** multiplied by advantage

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \mathbb{E}_{\hat{y}_t \sim \pi_{\theta}(\cdot | x, y_{<t})} \left[\overset{\text{scalar value}}{A(\hat{y}_t)} \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t | x, y_{<t}) \right]$$

$$A(\hat{y}_t) = \log \frac{\pi_{\theta}(\hat{y}_t | x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t | x, y_{<t})} \quad \begin{array}{l} \text{self-teacher} \\ \text{student} \end{array}$$

Method

- SDPO Loss: Reverse KL Divergence
 - Advantage provides **self-teacher's preference** (log) ratio

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \mathbb{E}_{\hat{y}_t \sim \pi_{\theta}(\cdot | x, y_{<t})} \left[\overset{\text{scalar value}}{A(\hat{y}_t)} \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t | x, y_{<t}) \right]$$

$$A(\hat{y}_t) = \begin{cases} 0 & \pi_{\theta}(\hat{y}_t | x, f, y_{<t}) = \pi_{\theta}(\hat{y}_t | x, y_{<t}) \\ + & \pi_{\theta}(\hat{y}_t | x, f, y_{<t}) > \pi_{\theta}(\hat{y}_t | x, y_{<t}) \\ - & \pi_{\theta}(\hat{y}_t | x, f, y_{<t}) < \pi_{\theta}(\hat{y}_t | x, y_{<t}) \end{cases} \quad \overset{\text{self-teacher}}{\quad} \overset{\text{student}}{\quad} \quad A(\hat{y}_t) = \log \frac{\pi_{\theta}(\hat{y}_t | x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t | x, y_{<t})}$$

Method

- SDPO Loss: Reverse KL Divergence
 - Advantage provides self-teacher's preference (log) ratio
 - This might bring **reasonable exploration!**
 - Fail -> feedback -> self-teacher's guidance -> explore -> ... -> success!

$$A(\hat{y}_t) = \begin{cases} 0 & \text{self-teacher } \pi_{\theta}(\hat{y}_t \mid x, f, y_{<t}) = \text{student } \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \\ + & \pi_{\theta}(\hat{y}_t \mid x, f, y_{<t}) > \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \\ - & \pi_{\theta}(\hat{y}_t \mid x, f, y_{<t}) < \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \end{cases}$$

Experiment 1

- Correct-solution strategy diversity
 - SDPO **explicitly exploits successful group rollout** as a feedback, collapsing its strategy

Table 3: Correct-Solution Strategy Diversity. (Judge: GPT-5.4-mini).

| Model | Pass@16 (%) | Avg. Strategies | Effective Strategies | Single-Strategy (%) |
|----------------|-------------|-----------------|----------------------|---------------------|
| Base (Qwen 8B) | <u>42.7</u> | <u>1.66</u> | <u>1.44</u> | <u>53.6</u> |
| SDPO | 49.6 | 1.15 | 1.10 | 90.8 |

Experiment 1

- Correct-solution strategy diversity (PoC)
 - SDPO explicitly exploits successful group rollout as feedback, collapsing its strategy
 - While the *failure-only* version **explores and nails diverse strategies**, even exceeding the base model's diversity

Table 3: Correct-Solution Strategy Diversity. (Judge: GPT-5.4-mini).

| Model | Pass@16 (%) | Avg. Strategies | Effective Strategies | Single-Strategy (%) |
|----------------|-------------|-----------------|----------------------|---------------------|
| Base (Qwen 8B) | <u>42.7</u> | <u>1.66</u> | <u>1.44</u> | <u>53.6</u> |
| SDPO | 49.6 | 1.15 | 1.10 | 90.8 |
| Failure only | 43.5 | 1.86 | 1.64 | 45.6 |

Experiment 1

- Correct-solution strategy diversity
 - However, we would like to increase our sampling efficiency as well
 - In other words, we need to **reward successful rollouts**

Table 3: Correct-Solution Strategy Diversity. (Judge: GPT-5.4-mini).

| Model | Pass@16 (%) | Avg. Strategies | Effective Strategies | Single-Strategy (%) |
|----------------|-------------|-----------------|----------------------|---------------------|
| Base (Qwen 8B) | <u>42.7</u> | <u>1.66</u> | <u>1.44</u> | <u>53.6</u> |
| SDPO | 49.6 | 1.15 | 1.10 | 90.8 |
| Failure only | 43.5 | 1.86 | 1.64 | 45.6 |

Method

- Forward KL Divergence
 - The advantage is always **equal or greater than zero**
 - But, the value is still control by the teacher's preference ratio

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \mathbb{E}_{\hat{y}_t \sim \pi_{\theta}(\cdot | x, y_{<t})} \left[A(\hat{y}_t) \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t | x, y_{<t}) \right]$$

$$A(\hat{y}_t) = \frac{\pi_{\theta}(\hat{y}_t | x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t | x, y_{<t})}$$

Method

- Forward KL Divergence
 - The advantage is always equal or greater than zero
 - Perfect match for **successful rollout/distribution**
 - Can we utilize this inductive bias of the loss?

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \mathbb{E}_{\hat{y}_t \sim \pi_{\theta}(\cdot | x, y_{<t})} \left[A(\hat{y}_t) \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t | x, y_{<t}) \right]$$

$$A(\hat{y}_t) = \frac{\pi_{\theta}(\hat{y}_t | x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t | x, y_{<t})}$$

Experiment 2

- **Routing** rollouts based on their result
 - Successful -> Forward KL: non-negative advantage
 - Failure -> Reverse KL: negative to positive, based on self-teacher
 - Combine each loss with equal weight (0.5)

$$A^{\text{FKL}}(\hat{y}_t) = \frac{\pi_{\theta}(\hat{y}_t \mid x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})}$$

self-teacher
student

$$A^{\text{RKL}}(\hat{y}_t) = \log \frac{\pi_{\theta}(\hat{y}_t \mid x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})}$$

self-teacher
student

Experiment 2

- Routing rollouts based on their result
(Successful -> Forward KL, Failure -> Reverse KL)
 - Intermediate levels of diversity and performance compared to the Base and SDPO

Table 4: Correct-Solution Strategy Diversity. (Judge: GPT-5.4-mini)

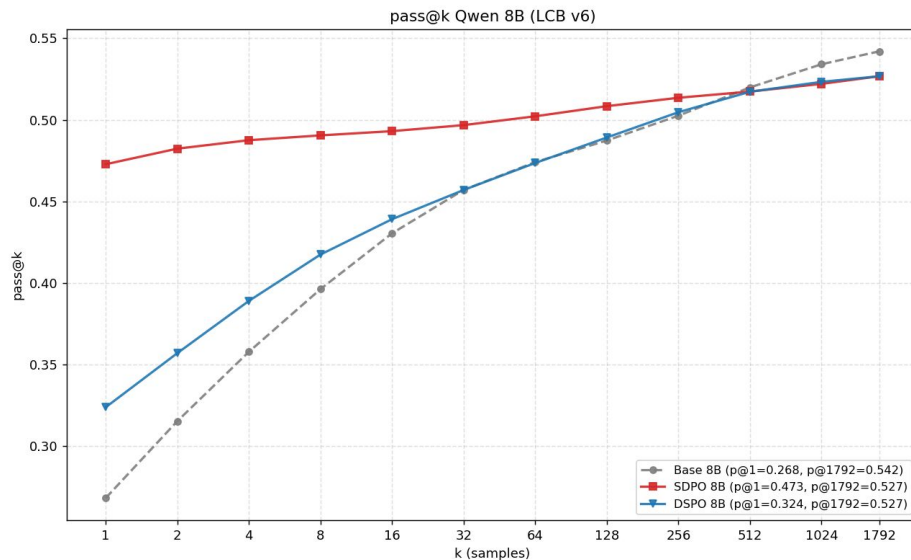
| Model | Pass@16 (%) | Avg. Strategies | Effective Strategies | Single-Strategy (%) |
|----------------|-------------|-----------------|----------------------|---------------------|
| Base (Qwen 8B) | 42.7 | 2.05 | 1.70 | 41.1 |
| SDPO | 49.6 | 1.03 | 1.02 | 96.9 |
| Ours | <u>45.0</u> | <u>1.63</u> | <u>1.43</u> | <u>61.0</u> |

Experiment 2

- Routing rollouts based on their result

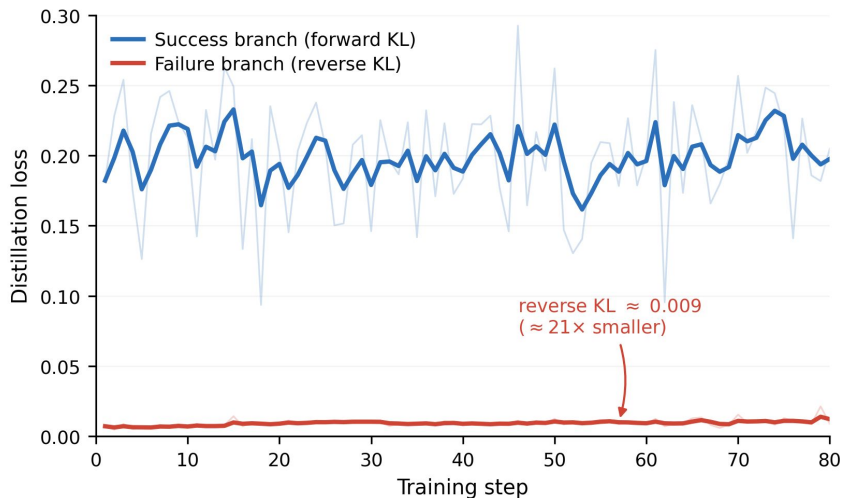
(Successful -> Forward KL, Failure -> Reverse KL)

- While we match SDPO's pass@k for larger k, we failed to expand coverage from the Base model
 - i.e., we failed to encourage model to explore diverse strategy



Discussions

- Loss perspective: Forward KL (success branch) dominates the loss
 - Self-teacher conditioning on its own successful rollout is **extremely peaked**
 - Combining losses with identical weights led to an imbalance in learning signals



Self-teacher peakedness: success vs. fail

| Branch | Distribution | $p_{\text{top-1}}$ | H (nats) | top-1 disagree |
|---------|--------------|-----------------------|-----------------------|----------------|
| Success | Student | 0.884 (± 0.187) | 0.316 (± 0.479) | 7.0% |
| | Self-teacher | 0.984 (± 0.065) | 0.050 (± 0.172) | |
| Fail | Student | 0.834 (± 0.219) | 0.470 (± 0.591) | 1.9% |
| | Self-teacher | 0.834 (± 0.219) | 0.470 (± 0.591) | |

Discussions

- Divergence direction perspective
 - RKL (failed rollout) is inherently unable to quickly boost the probability of student's **low-probability actions**

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \sum_{\hat{y}_t}^{|V|} \overset{\text{student}}{\boxed{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})}} \left[A(\hat{y}_t) \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \right]$$

dilutes the advantage

$$A^{\text{RKL}}(\hat{y}_t) = \log \frac{\pi_{\theta}(\hat{y}_t \mid x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})}$$

Discussions

- Divergence direction perspective
 - This issue is further exacerbated when applying top-K approximation
 - Self-teacher might prefer some actions, but it doesn't count in the student top-K
 - In general, a model failure indicates that **high-quality actions are concealed** under low probabilities in the model distribution.
 - Therefore, it is unsuitable for exploring novel actions

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \sum_{\hat{y}_t}^{|V|} \overset{\text{student}}{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})} \left[A(\hat{y}_t) \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \right]$$

$$A^{\text{RKL}}(\hat{y}_t) = \log \frac{\pi_{\theta}(\hat{y}_t \mid x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})}$$

Discussions

- Divergence direction perspective
 - In contrast, FKL functions **regardless of the student model's own probability distribution**
 - We can efficiently boost the probabilities of actions that self-teacher assigns high probability (which is included in teacher's top-K)

$$\begin{aligned}\nabla_{\theta} J &= \sum_{t=1}^{|y|} \mathbb{E}_{\hat{y}_t \sim \pi_{\theta}(\cdot | x, y_{<t})} \left[\frac{\pi_{\theta}(\hat{y}_t | x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t | x, y_{<t})} \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t | x, y_{<t}) \right] \\ &\quad \text{self-teacher} \\ &= \sum_{t=1}^{|y|} \sum_{\hat{y}_t}^{|V|} \boxed{\pi_{\theta}(\hat{y}_t | x, f, y_{<t})} \left[\nabla_{\theta} \log \pi_{\theta}(\hat{y}_t | x, y_{<t}) \right]\end{aligned}$$

Discussions

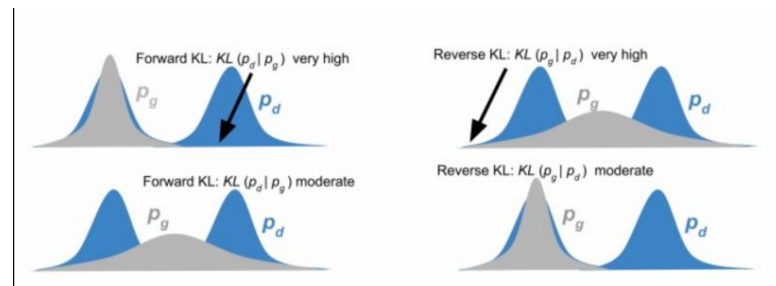
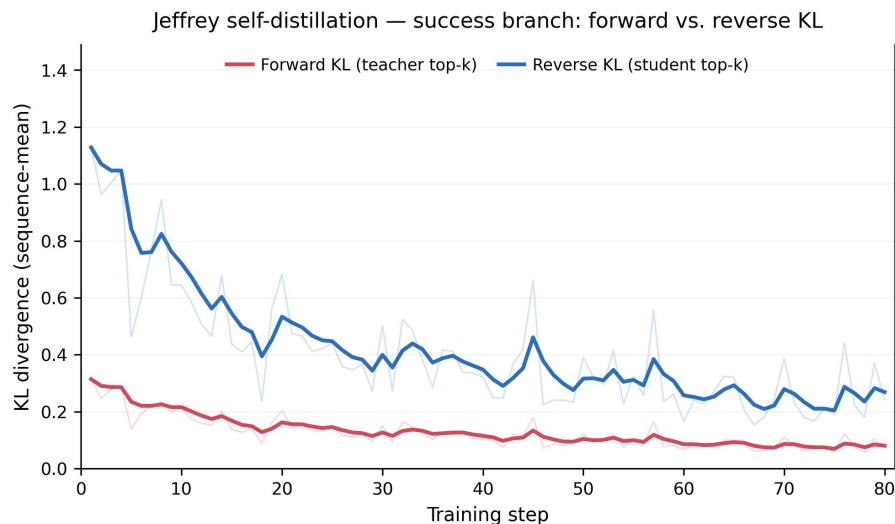
- Divergence direction perspective
 - Conversely, RKL can be more appropriate for successful rollouts.
 - The **success** of a rollout means the model has **already allocated decent probabilities** to the respective actions
 - There is no need to boost new actions; we merely need to slightly refine the current distribution (mode seeking)

$$\nabla_{\theta} J = \sum_{t=1}^{|y|} \sum_{\hat{y}_t}^{|V|} \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \left[A(\hat{y}_t) \cdot \nabla_{\theta} \log \pi_{\theta}(\hat{y}_t \mid x, y_{<t}) \right]$$

$$A^{\text{RKL}}(\hat{y}_t) = \log \frac{\pi_{\theta}(\hat{y}_t \mid x, f, y_{<t})}{\pi_{\theta}(\hat{y}_t \mid x, y_{<t})}$$

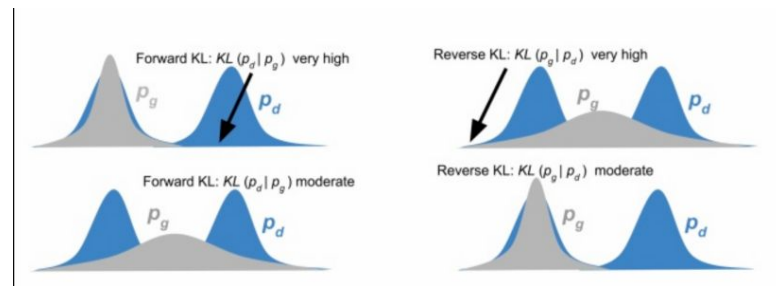
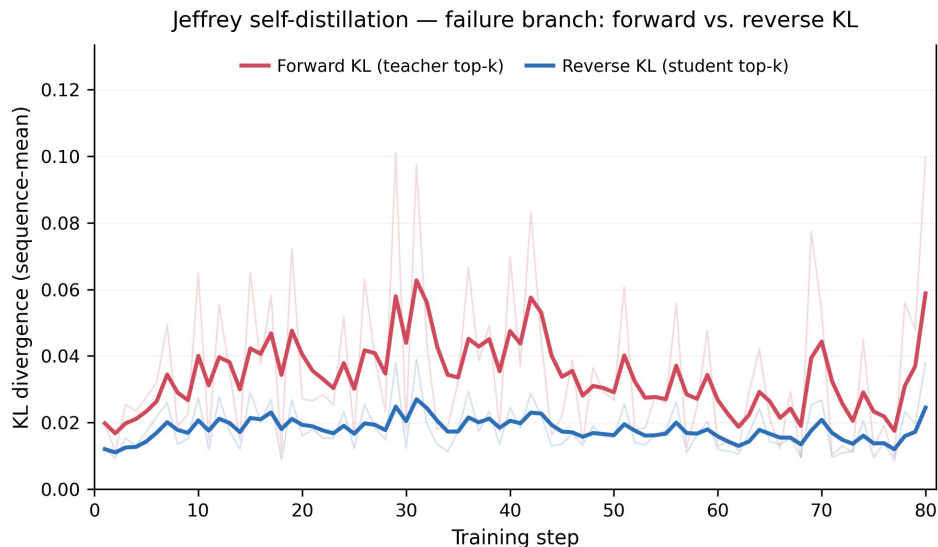
Discussions

- Reverse KL is larger than Forward KL in Success Branch
 - In the success branch, RKL dominates FKL by focusing on **refining already decent-probability actions**
 - FKL is low because the support of the distributions are similar



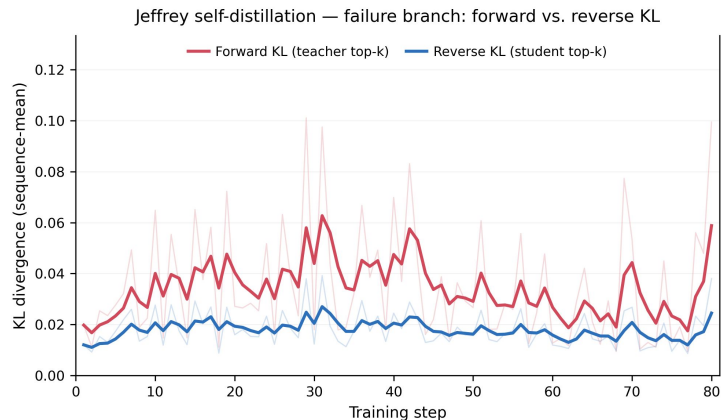
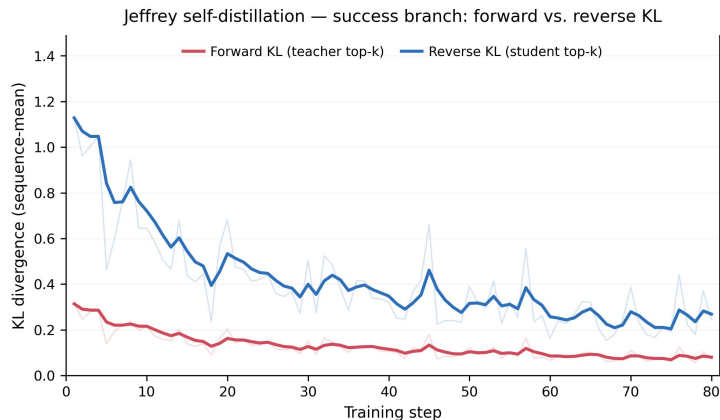
Discussions

- Forward KL is larger than Reverse KL in Failure Branch
 - This means the mass-covering phenomenon exists within failure branches, where the **teacher favors student's low-probability actions**
 - However, RKL still can be useful for some cases



Discussions

- Routing is therefore important
 - Each branch possesses different problem; it **requires different loss function**
 - This does not mean each branch should be restricted to a single KL direction's loss

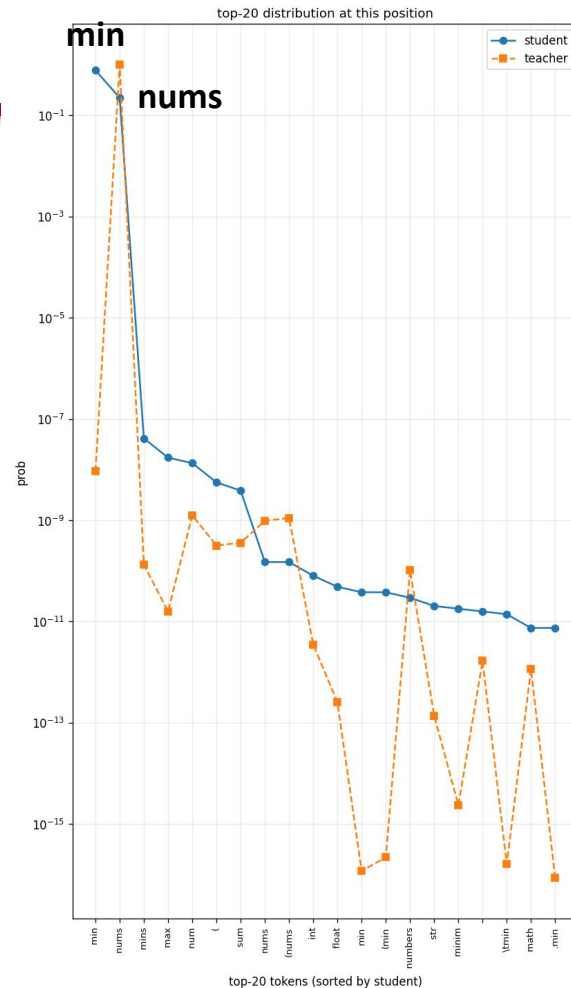


Discussions

- Qualitative results (success)
 - Student
 - Top 1: min, Top 2: nums
 - Samples “nums”, get correct
 - Self-teacher
 - Prefers “nums”
 - RKL: 13.85 >> FKL: 1.5

```
i in range(len(nums) - 1):
    current_sum = nums[i] + nums[i + 1]
    if current_sum < min_sum:
        min_sum = current_sum
        min_index = i

# Replace the pair with their sum
nums[min_index] = nums[min_index] + nums[min_index + 1]
nums.pop(min_index + 1)
operations += 1
```



Discussions

- Qualitative results (failed)
 - Student
 - Gets wrong
 - Receives test case “KSHJFGF”
 - Self-teacher
 - Assigns high probability “K” in examples
 - FKL = 6.22 >> RKL = 0.48

==== self-teacher injected: FEEDBACK ====

Test Case 21: Wrong Answer

Input
KSHJFGF

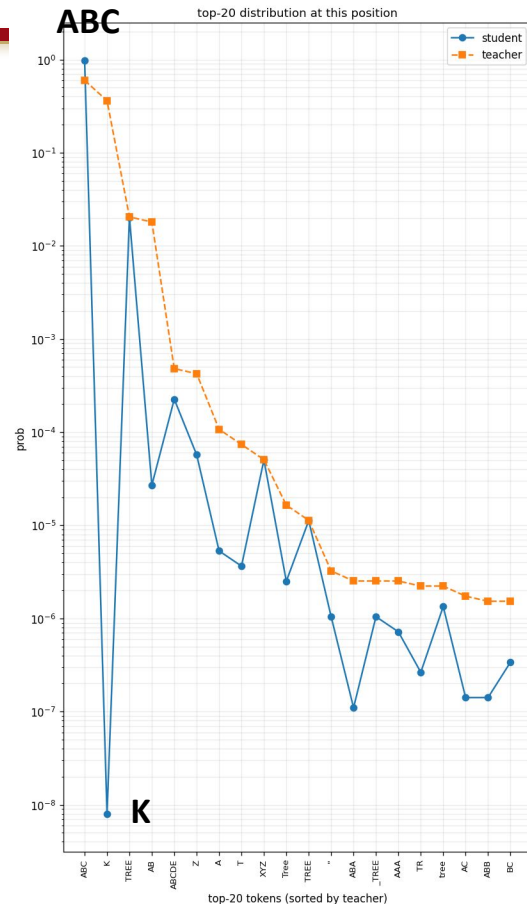
Output
KSHJFGGFJHJK

Expected
KSHJFGFJHJK

Test Case 24: Wrong Answer
...(truncated)

highlight = divergence token

Example:
- Input: `ABC`
- Output: `ABCBA`
- Input: `Z`
- Output: `Z`
- Input: `TREE`
...



Conclusion and Limitations

- While **learning from failure leads to diverse** strategies, how to harmoniously **combine it with learning from success** remains an open question
 - Gradient manipulation techniques, such as PCGrad, could be a promising solution to resolve this conflict
- Although we showed which **KL direction is crucial depending on the outcome**, switching between them creates a **discontinuous optimization landscape**
 - For instance, using FKL to boost probabilities eventually leads to success, forcing a sudden shift to RKL. This discontinuity links back to the challenge of balancing success and failure
- We argued that pursuing **diverse strategies maintains a high pass@k** even at large k , but we could not verify this empirically.
 - However, recent results from Poly-EPO strongly support the validity of our claim

Thank you

Q&A



Author Contributions

- **Ahn** led the team, proposed core algorithmic ideas
- **Moon** led the experiments and evaluation of the project
- **Jeong** led the writing of the proposal, contributed to the project direction in discussions